FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 3.3: VECTORS I	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

Your mission: Write a Final Explanation that answers the driving question — "Why did the Kisumu ferry miss Mbita, and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind?" — using everything you have learned about vectors across this unit. Your explanation should be clear, complete, and mathematically precise. Use diagrams, vector notation, calculations, and real examples from the ferry phenomenon and other contexts. A strong Final Explanation does three things: (1) it explains the science and mathematics behind the phenomenon, (2) it uses correct vocabulary and notation throughout, and (3) it shows how the mathematical tools you learned can solve real-world navigation problems. Write your explanation in full paragraphs and include at least ONE clearly labelled vector diagram and ONE worked calculation. Your target audience is a fellow Grade 10 student who missed this unit — explain it so clearly that they could understand the ferry situation and use vectors confidently.

Section 1: Why the Ferry Missed — Scalars vs. Vectors

Explain why the Kisumu ferry missed Mbita even though the captain pointed the engine directly east. In your answer: (a) define scalar and vector quantities and give at least THREE examples of each from everyday Kenyan life, (b) explain why the captain's speed reading alone was not enough information to predict where the ferry would land, and (c) use the words magnitude and direction correctly in your explanation.	The Kisumu ferry missed Mbita because the captain only controlled one of the two directed quantities acting on the vessel. To understand why, we must first distinguish between scalars and vectors. A scalar quantity has magnitude only — it tells you how much but not which way. Examples from everyday Kenyan life include the temperature of water boiling for ugali (100 °C), the mass of a sack of maize at Kisumu market (90 kg), and the time it takes to travel from Nairobi to Mombasa by SGR (approximately 4.5 hours). A vector quantity, by contrast, has both magnitude AND direction — it tells you how much and which way. Examples include the displacement of a matatu travelling 15 km north from Kisumu town centre, the velocity of wind blowing south-west across Lake Victoria at 8 km/h, and the force a porter exerts pushing a trolley eastward at Mombasa port. The captain's engine speed of 20 km/h is, on its own, a scalar — it tells us how fast but not the total effect on the ferry's journey. The moment the southward lake current of 8 km/h also acted on the vessel, there were TWO vectors present simultaneously: the engine velocity (20 km/h due east) and the current velocity (8 km/h due south). Because both vectors acted at the same time, the ferry did not travel purely east — it was continuously pushed sideways as well. The captain's dashboard speed reading captured only the engine's contribution and gave no information about the current's direction. This is why direction is essential: ignoring it meant ignoring the very force that caused the ferry to drift. Scalars cannot capture this situation; only vectors can.

Section 2: Representing Vectors — Notation, Diagrams, and the Number Line	
Show how vectors can be written and drawn. In your answer: (a) write the engine velocity and the current velocity of the Kisumu ferry using correct vector notation (bold or arrow notation, column vectors, and bearing or compass direction), (b) draw and label a clear vector diagram showing both vectors as arrows with correct relative lengths and directions, and (c) explain what the tail, head, magnitude, and direction of a vector arrow each represent.	Before we can solve the ferry's navigation problem mathematically, we must be able to write and draw vectors clearly. There are several standard ways to represent a vector. Using arrow (bold) notation, we write the engine velocity as **v** = 20 km/h due east, and the current as **c** = 8 km/h due south. Using column vector notation, we place east as the positive x-direction and north as the positive y-direction, so: Engine velocity: **v** = (20, 0) [20 km/h east, 0 km/h north/south] Current velocity: **c** = (0, −8) [0 km/h east/west, 8 km/h south] Using bearing notation, the engine points on a bearing of 090° (due east) and the current flows on a bearing of 180° (due south). In a vector diagram, each vector is drawn as a directed arrow. The TAIL is the starting point of the arrow — where the push begins. The HEAD (arrowhead) shows the direction the push acts. The LENGTH of the arrow represents the magnitude of the vector, drawn to scale (for example, 1 cm = 4 km/h). The angle the arrow makes with north or east shows the direction. [DIAGRAM: Draw a horizontal arrow of length 5 cm pointing right, labelled 'Engine velocity: 20 km/h East'. Below it, draw a vertical arrow of length 2 cm pointing downward, labelled 'Current: 8 km/h South'. Label the tail of each arrow with a dot and the head with an arrowhead. Include a compass rose in the top-right corner and a scale bar: 1 cm = 4 km/h.] Representing vectors this way makes it impossible to confuse the engine push with the current push — they are visibly different in both length and direction. This precision is what mathematics offers the captain that a simple speed dial cannot.

Section 3: Adding Vectors — Finding the Ferry's True Journey	
Explain and demonstrate vector addition using the ferry scenario. In your answer: (a) state the triangle law and parallelogram law of vector addition, (b) use the tip-to-tail method to find the resultant velocity of the ferry as a column vector, (c) calculate the magnitude (actual speed) of the resultant vector using Pythagoras' theorem, (d) calculate the true direction of travel using trigonometry (tan ratio), and (e) explain what the resultant vector tells the captain about where the ferry	Vector addition is the mathematical tool that explains exactly where the ferry ended up. When two or more vectors act simultaneously on an object, they combine into a single RESULTANT vector that has the same effect as all the individual vectors together. The TRIANGLE LAW states: place the tail of the second vector at the head of the first; the resultant is the vector drawn from the tail of the first to the head of the second. The PARALLELOGRAM LAW states: place both vectors tail-to-tail; complete the parallelogram; the diagonal from the common tail is the resultant. Using the tip-to-tail (triangle) method for the ferry: Engine velocity: **v** = (20, 0) Current velocity: **c** = (0, −8) Resultant velocity: **R** = **v** + **c** = (20 + 0, 0 + (−8)) = (20, −8)

actually went.	This resultant tells us the ferry moved 20 km/h east AND 8 km/h south simultaneously — a combined motion in a south-easterly direction. Calculating the MAGNITUDE (actual speed) using Pythagoras' theorem: |**R**| = $\sqrt{(20^2 + 8^2)}$ = $\sqrt{(400 + 64)}$ = $\sqrt{464}$ ≈ 21.5 km/h So the ferry actually travelled at approximately 21.5 km/h — faster than the engine alone, because the current added a southward component. Calculating the TRUE DIRECTION using trigonometry: $\tan \theta$ = opposite/adjacent = 8/20 = 0.4 $\theta = \tan^{-1}(0.4) \approx 21.8°$ This means the ferry's true path was approximately 21.8° south of due east — a bearing of about 111.8°. Instead of landing at Mbita (due east), it landed roughly 21.8° off course to the south. [DIAGRAM: Draw the engine vector (20, 0) as a long horizontal arrow east. From its head, draw the current vector (0, −8) as a shorter arrow pointing south. Draw the resultant from the original tail to the new head. Label all three sides: '20 km/h East', '8 km/h South', 'Resultant ≈ 21.5 km/h at 21.8° S of E'. Mark the angle θ at the starting point.] This is the precise mathematical answer to why the ferry missed Mbita: the resultant of the two simultaneous vectors carried the vessel south-east, not east.

Section 4: Vector Properties — What the Captain Must Understand	
Explain the key properties of vectors that are essential for the captain to navigate correctly. In your answer: (a) explain why the ORDER of vector addition does not change the resultant (commutativity), using the ferry vectors as proof, (b) explain why DIRECTION of each individual vector matters enormously — what would happen if the current were north instead of south? (c)	For the captain to use vectors as a reliable navigation tool, there are four key properties to understand deeply. First: VECTOR ADDITION IS COMMUTATIVE — the order does not matter. Whether you add the engine vector first and then the current, or the current first and then the engine, the resultant is identical: **v** + **c** = (20, 0) + (0, −8) = (20, −8) **c** + **v** = (0, −8) + (20, 0) = (20, −8) ✓ This is reassuring: it does not matter whether the current started before the engine or after — the combined effect is always the same resultant displacement. Second: DIRECTION OF EACH VECTOR MATTERS ENORMOUSLY. If the current were 8 km/h DUE NORTH instead of south, the resultant would be **R** = (20, 0) + (0, 8) = (20, 8) — sending the ferry north-east of Mbita instead of south-east. The magnitude would still be ≈ 21.5 km/h, but the ferry would miss Mbita in the completely opposite direction! This shows that two

define and give an example of a zero vector and an equal vector in the context of the ferry, and (d) explain what a negative vector means and how the captain could use it.	vectors with the same magnitude but different directions are entirely different vectors. A Kericho tea farmer facing a wind blowing toward Kisumu experiences something completely different from one facing wind blowing away from Kisumu — even if both winds are the same speed. Third: A ZERO VECTOR has magnitude 0 and no defined direction, written as **0** = (0, 0). If there were no current at all on Lake Victoria, **c** = (0, 0), and the resultant would equal the engine velocity exactly — the ferry would reach Mbita perfectly. Equal vectors have the same magnitude AND same direction; if another ferry on the lake has **v** = (20, 0), it is the equal vector to the Kisumu ferry's engine velocity, regardless of where on the lake it is. Fourth: A NEGATIVE VECTOR has the same magnitude but exactly opposite direction. The negative of the current **c** = (0, −8) is −**c** = (0, 8) — a northward current of 8 km/h. If the captain could generate a thrust of (0, 8) northward in addition to the engine, these two would cancel: **c** + (−**c**) = **0**, eliminating the drift entirely. This is precisely the strategy the captain needs — understanding negative vectors is the first step toward correcting course.

Section 5: Giving the Captain the Tools — Solving the Navigation Problem	
Use everything you have learned to solve the ferry's real navigation problem. In your answer: (a) calculate the actual displacement of the ferry (how far south of Mbita it landed) given the crossing time for 60 km at the resultant speed, (b) explain what heading (direction) the captain should set the engine BEFORE departing so that the resultant of engine thrust and current lands the ferry exactly at Mbita, (c) show this corrected vector calculation fully, and (d) write a clear recommendation to the Kenya Ferry Services captain — in your own words — explaining what vectors can do that a simple speed dial cannot, and why every navigator on Lake Victoria should understand resultant vectors.	Now we can give the captain the mathematical tools to always reach Mbita correctly. PART A — How far south did the ferry land? The ferry needed to cross 60 km eastward. At the engine's eastward component of 20 km/h, the crossing took: Time = distance ÷ speed = 60 ÷ 20 = 3 hours During those 3 hours, the southward current of 8 km/h pushed the ferry: Southward drift = 8 × 3 = 24 km south of Mbita So the ferry landed 24 km south of its intended destination — a significant and dangerous miss that explains the delay and confusion for hundreds of passengers and traders. PART B — What heading should the captain use next time? To cancel the 8 km/h southward current, the captain must aim the engine slightly NORTH of east, so that the northward component of the engine velocity exactly cancels the southward current. Let the engine speed remain 20 km/h and let angle α be the angle north of east. For the northward component to cancel the current: $20 \times \sin \alpha = 8$ $\sin \alpha = 8/20 = 0.4$ $\alpha = \sin^{-1}(0.4) \approx 23.6°$ north of east (i.e., a bearing of approximately 066.4°) The effective eastward speed then becomes: Eastward component = $20 \times \cos(23.6°) = 20 \times 0.917 \approx 18.3$ km/h

	Resultant with current: **R** = (18.3, 8) + (0, −8) = (18.3, 0) ✓ The ferry now travels due east at 18.3 km/h and will land exactly at Mbita — slightly slower than before, but precisely on target. New crossing time = 60 ÷ 18.3 ≈ 3.28 hours (about 3 hours 17 minutes) — a small time cost for arriving at the correct destination. [DIAGRAM: Draw the corrected engine vector at angle 23.6° north of east. Draw the southward current vector from its head. Show the resultant as a horizontal arrow pointing due east. Label all angles, magnitudes, and directions.] RECOMMENDATION TO THE CAPTAIN: Dear Captain, a speed dial tells you how hard your engine is pushing — but it cannot tell you where you will end up, because it ignores every other force acting on your vessel. Vectors give you the complete picture: magnitude AND direction for every force. By calculating the resultant of your engine thrust and the lake current BEFORE you depart, you can set a corrected heading that guarantees arrival at Mbita every single time, regardless of how strong or which way the current flows. Mathematics does not replace your experience — it sharpens it, turning guesswork into certainty. Every navigator on Lake Victoria who understands resultant vectors carries a tool more reliable than any instrument on the dashboard.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding of Vectors vs. Scalars	Clearly and accurately distinguishes vectors from scalars with precise definitions. Provides three or more correct, Kenya-relevant examples of each. Correctly uses the terms magnitude and direction throughout, and explicitly links the distinction to why the ferry missed Mbita.	Correctly defines vectors and scalars and gives at least two examples of each. Uses magnitude and direction appropriately in most of the explanation. Connection to the ferry phenomenon is present but may lack depth.	Attempts to distinguish vectors from scalars but definitions are incomplete or contain errors. Fewer than two examples are given, or examples are incorrect. The link between the scalar/vector distinction and the ferry phenomenon is unclear or missing.
Vector Representation and Notation	Uses at least two correct forms of vector notation (e.g., column vectors, bold/arrow notation, bearings). Draws a clearly labelled vector diagram with correct relative lengths, directions, arrowheads, a scale, and a compass reference. All notation is consistent and error-free.	Uses one correct form of vector notation and draws a diagram that is mostly accurate. The diagram includes labels and arrowheads but may be missing a scale or compass reference. Minor notation inconsistencies are present.	Attempts vector notation or a diagram but with significant errors in direction, length, or labelling. Vector arrows may be missing arrowheads or labels. Notation is inconsistent or confused with scalar notation.
Vector Addition and Resultant Calculation	Correctly states and applies at least one law of vector addition (triangle or parallelogram). Adds the ferry vectors accurately using column vectors.	Correctly adds the ferry vectors and obtains the right resultant column vector. Applies Pythagoras' theorem with mostly correct working. May make a minor	Attempts vector addition but makes errors in combining components (e.g., adds magnitudes as scalars). Attempts Pythagoras' theorem but sets it up

	Calculates the resultant magnitude using Pythagoras' theorem with correct working and a correct answer (≈ 21.5 km/h). Calculates the true direction using tan ratio with correct working (≈ 21.8°).	arithmetic error in the magnitude or direction calculation but the method is sound.	incorrectly or reaches a wrong answer without showing the error in method. Direction calculation is missing or incorrect.
Application: Solving the Navigation Problem	Correctly calculates the southward drift (24 km) with full working. Correctly sets up and solves the corrected heading problem using trigonometry (sin ratio), finds α ≈ 23.6° north of east, and verifies the resultant is due east. Recommendation to the captain is clear, mathematically grounded, and written in accessible language.	Correctly calculates the drift or the corrected heading (but not necessarily both with full working). The approach to the corrected heading is mathematically sound even if a minor arithmetic error occurs. Recommendation is present and generally accurate.	Attempts the drift calculation or the corrected heading but with significant method errors (e.g., using the wrong trigonometric ratio, or not accounting for the current in the resultant). Recommendation to the captain is vague or missing a mathematical basis.
Mathematical Communication and Use of Vocabulary	Uses all key unit vocabulary correctly and consistently throughout: vector, scalar, magnitude, direction, resultant, displacement, velocity, column vector, bearing, triangle law / parallelogram law, zero vector, negative vector, equal vectors. Writing is clear, logically structured, and accessible to a fellow Grade 10 student. Diagrams and calculations are integrated into the explanation rather than added separately.	Uses most key vocabulary correctly with occasional omissions or minor misuse. The explanation is generally clear and logically ordered. Diagrams and calculations are present and mostly integrated into the written explanation.	Uses fewer than half of the key vocabulary terms, or uses several terms incorrectly. The explanation lacks clear structure and may be difficult for a fellow student to follow. Diagrams or calculations are absent or disconnected from the written explanation.